Integrated Pest Management

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Integrated Pest Management (A07263)

Short Title:	Integrated Pest Management		
Faculty	Science and Computing		
Department	Science		
Cluster	Agriculture		
Author	TWOODCOCK		
Credits	5	**Level:**	Advanced

Description of Module / Aims

The module is designed for students who have a basic knowledge of agricultural or horticultural cropping systems or silviculture. It takes a multidisciplinary approach to crop-pest relationships in agriculture, horticulture and forestry, with examples and case studies drawn from all three disciplines. The students will carry out a case study on some aspect of IPM based on a literature search. Students who wish to be added to the Department of Agriculture SUD database need to select this elective and complete a shadowing exercise upon completion of their studies.

Programmes

MGTH-0016	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	4 / 8 / E
MGTH-0016	BSc (Hons) in Land Management (in Agriculture) (WD_SBSAG_B)	1 / 2 / E
MGTH-0016	BSc (Hons) in Land Management (in Forestry) (WD_SBSFO_B)	1 / 2 / E
MGTH-0016	BSc (Hons) in Land Management (in Horticulture) (WD_SBSHO_B)	1 / 2 / E

Indicative Content

- Forecasting and economic thresholds
- Decision support systems for pest management
- Pest monitoring using on-site monitors (e.g. splashometers re disease spores), diagnostic kits (e.g. immunoassay), pheromone traps
- Resistant crop varieties/cultivars, mixed cultivars, transgenic crops
- Cultural control (rotation, mechanical, undersowing, cover crops, GM)
- Biological control (beneficial predators, microbes Bt)
- Reduced and targeted use of selective pesticides

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Evaluate the contribution that IPM currently makes in the management of pests, and how this contribution could be extended further.
2. Critique the ecology of crop-pest relationships and infer how a multidisciplinary approach can be taken to pest management.
3. Recommend what IPM measures are appropriate to use in particular pest situations.
4. Assemble and assess data relevant to pest management.
5. Design an integrated approach to dealing with a selected crop-pest scenario.

Learning and Teaching Methods

- Lectures.
- Case studies.
- Field trips.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	70%	1,2,3
Continuous Assessment	30%	
Project	30%	4,5

Assessment Criteria

<40%: Very little knowledge and understanding of the subject matter. Unable to carry out critical thinking or practical skills.

40%-49%: Will be able to show a good knowledge of the key learning outcomes, including the contribution that IPM makes in the management of pests. The student will be able to write and present a case study.

50%-59%: As well as the above, the student will be able to demonstrate a good understanding of crop-pest relationships and a multidisciplinary approach to pest management. The student will be able to write and present a quality case study.

60%-69%: In addition, the student will demonstrate more detailed knowledge of all the topics, including how to determine what IPM measures to use in particular situations. The student will also be able to write and present a good quality case study.

70%-100%: The learner will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, including the role of pest management and be able to write and present a high quality case study.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Field Trips	12	
Independent Learning	87	

Essential Material

- Ellis , S., S. White, J. Holland, B. Smith, R. Collier and A. Jukes. _Encyclopaedia of pests and natural enemies in field crops_. UK: Agriculture & Horticulture Development Board, 2014.
- Heyler, N., K. Brown and N. Cattlin. _Biological Control in Plant Prote__ction_. London: Manson Publishing Ltd, 2003.
- Maredia, K.D., D. Dakouo and D. Mota-Sanchez. _Integrated Pest Management in the Global Arena_. UK: Wallingford, 2003.
- Radcliffe, E.B., W.D. Hutchison and R.E. Cancelado. _Integrated Pest Management: Concepts, Tactics, Strategies and Case Studies_. UK: Cambridge University Press, 2008 (Online 2010).

Supplementary Material

- "University of California Davis IPM Online." UC IPM Online. 2015. http://www.ipm.ucdavis.edu/